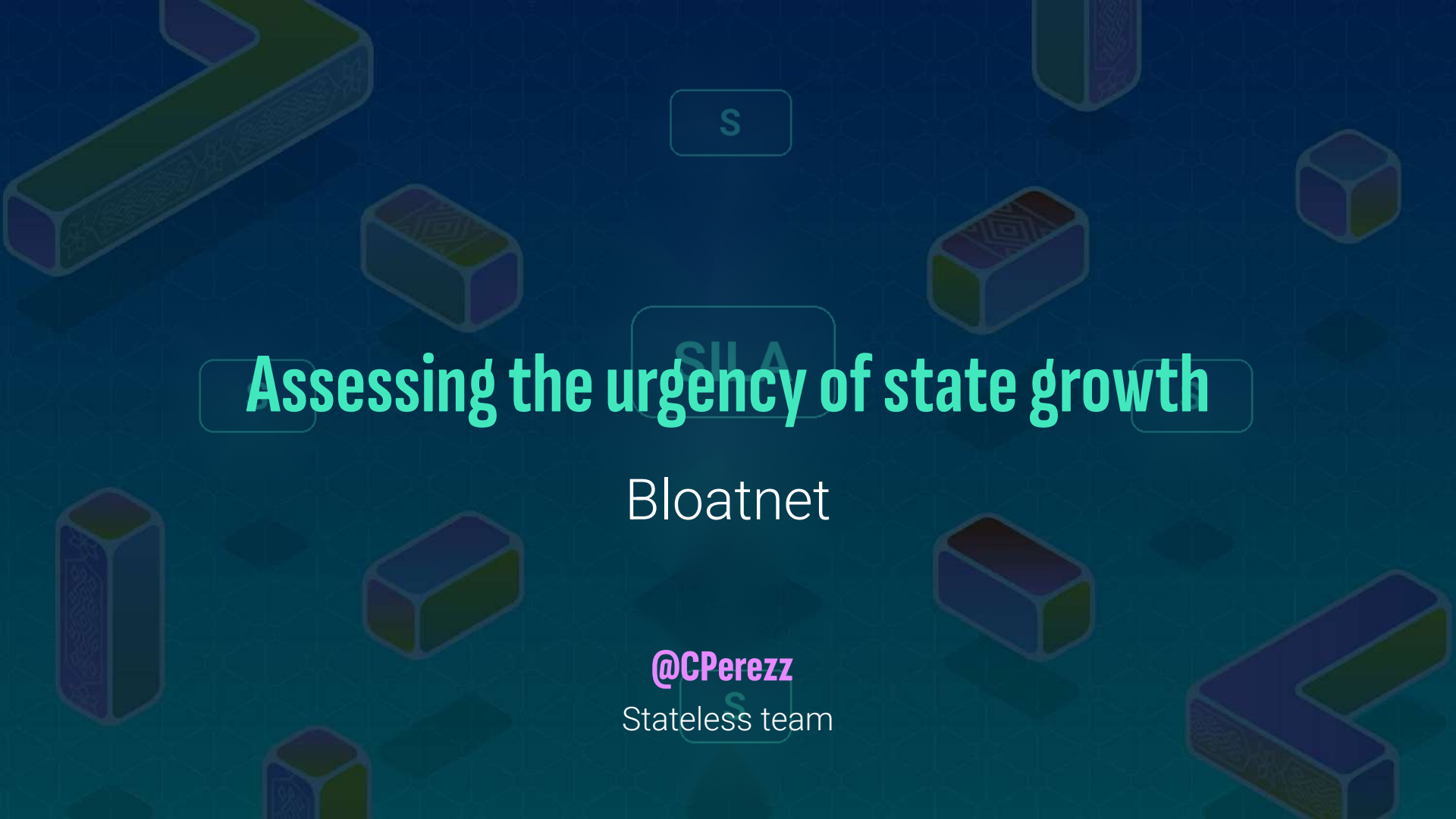


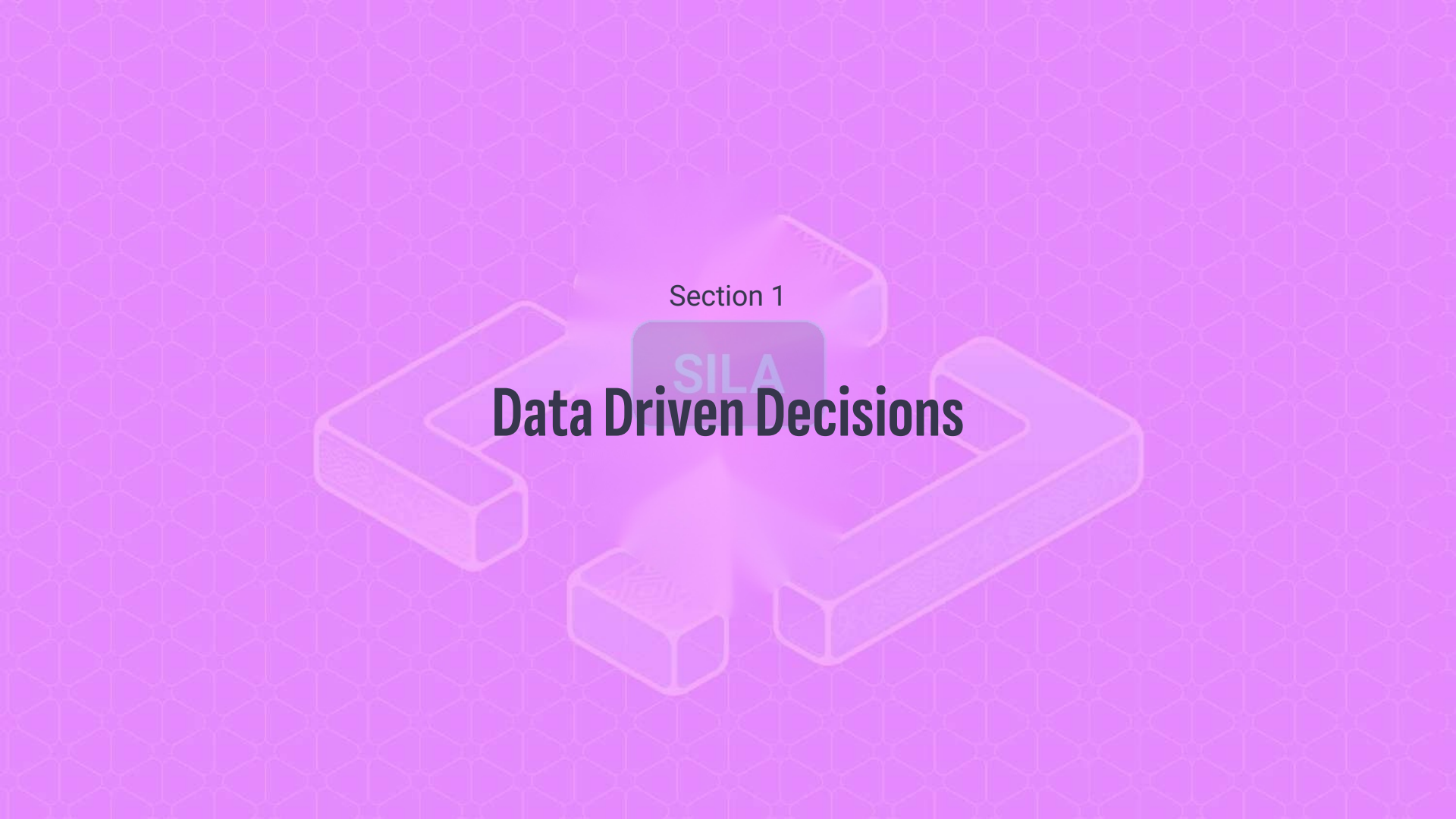


Assessing the urgency of state growth

Bloatnet

@CPerezz

Stateless team



Section 1

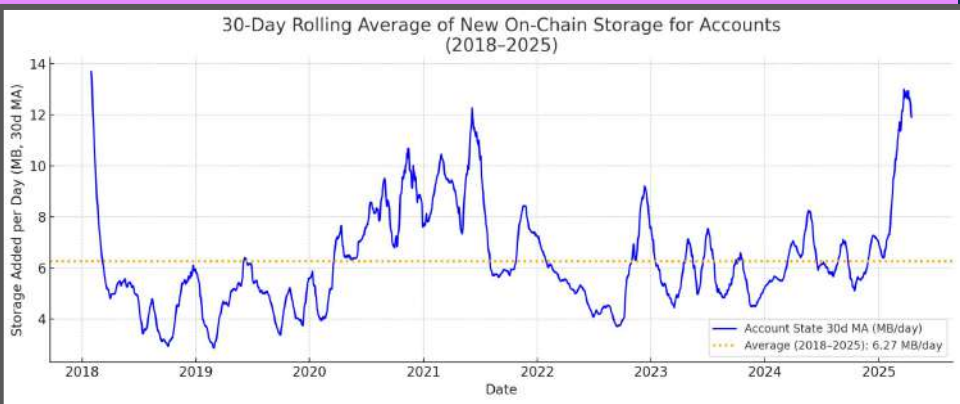
Data Driven Decisions

We can't be a stone on the path

We do care about statelessness. And we work everyday to make it happen. And the problems we address are eventually going to need a solution. **BUT:**

- Sila has lots of simultaneous priorities:
 - Throughput
 - UX
 - Data availability
- We can't go stale on these because of the “vibes” that state-growth and gas limit increases give us.

The word "SILA" is displayed in a bold, teal-colored, sans-serif font. It is centered within a rounded rectangular frame that has a thin teal border and a light teal gradient fill. The background of the slide features a subtle, repeating pattern of hexagons in a slightly darker shade of teal.



We can't predict the future

State growth has clearly a bounded overall maximum set by the gas limit of the network and the assumption that (we can't do more SSTORE/CREATE.. than gas limit we have available per block).

With gas limit increases, we get an uncertainty, **when and how will state growth <-> gas limit become a problem?**



Section 2

So let's collect the data, no?

Here's the key points for BloatNet

When do we need to prioritize statelessness?



Know when state growth becomes a critical issue. When nodes start really suffering to keep up. And when we can't solve it by optimizing the client code more.

Massive State testnet data



Gather extensive performance data with high gas limits. Try out how new SIP impls will behave with 10x state or do more precise research based on real data.

Clear the path for SIP-7938



Controversial or not, this SIP challenges sila to become competitive. We want to support a much easier and clear decision-making when it comes to it.

bloatnet.info



SCAN ME

Call for participation. SPAM00R scenarios!

Currently coding about 10-15 scenarios which combine:

- Bloating
- Attacks/node stressing
- Special interesting scenarios (XEN).

```
INFO[0183] Progress: sent 580 txs, deployed 504 contracts scenario=contract-deploy
INFO[0184] Progress: sent 590 txs, deployed 504 contracts scenario=contract-deploy
INFO[0185] Progress: sent 600 txs, deployed 528 contracts scenario=contract-deploy
INFO[0186] Block deployment summary avg_gas_per_byte=224.54 block_number=90 contracts_deployed=24 scenario=contract-deploy total_bytecode_size=528888 total_contracts=528 total_gas_used=118756776
INFO[0186] Progress: sent 610 txs, deployed 528 contracts scenario=contract-deploy
INFO[0188] Progress: sent 620 txs, deployed 528 contracts scenario=contract-deploy
```

Draft Add/state growth worst cases for BloatNet #41		
(CPerez wants to merge 37 commits into silendrop/master from CPerez/add/state_growth_worst_...)		
feat(extcodesize-overload): add new scenario for EXTCODESIZE overload...	Verified	63caffb
update: Contract bytecode reduction to account for extra costs	Verified	cd23bd4
feat(contract-deploy): improve bytecode size ratio & auto-adapt to ga...	Verified	99bc827
feat(setcodetx): support batching for max bloating mode to overcome m...	Verified	1657334
feat(setcodetx): fix funding and confirmation handling in max bloatin...	Verified	83a085d
chore(eao-delegation): Split eao delegation scenarios on bloating and...	Verified	05b216e
chore: revert changes to setcodetx	Verified	e3e32aa



CPerez commented 9 hours ago

Author ...

As mentioned in Gas Limit Testing group, ZEN contracts are certainly an interesting scenario/benchmark to have due to the fact that they are A REAL WORST CASE and exist on sila-mainnet nowadays.

Thus, will be adding scenarios to support the following:

- ☐ ZEN contract-like interactions (batchedMint in particular)
- ☐ MAX SSTORE in a block
- ☐ MAX CREATE2
- ☒ MAX CREATE

As mentioned by Storm:

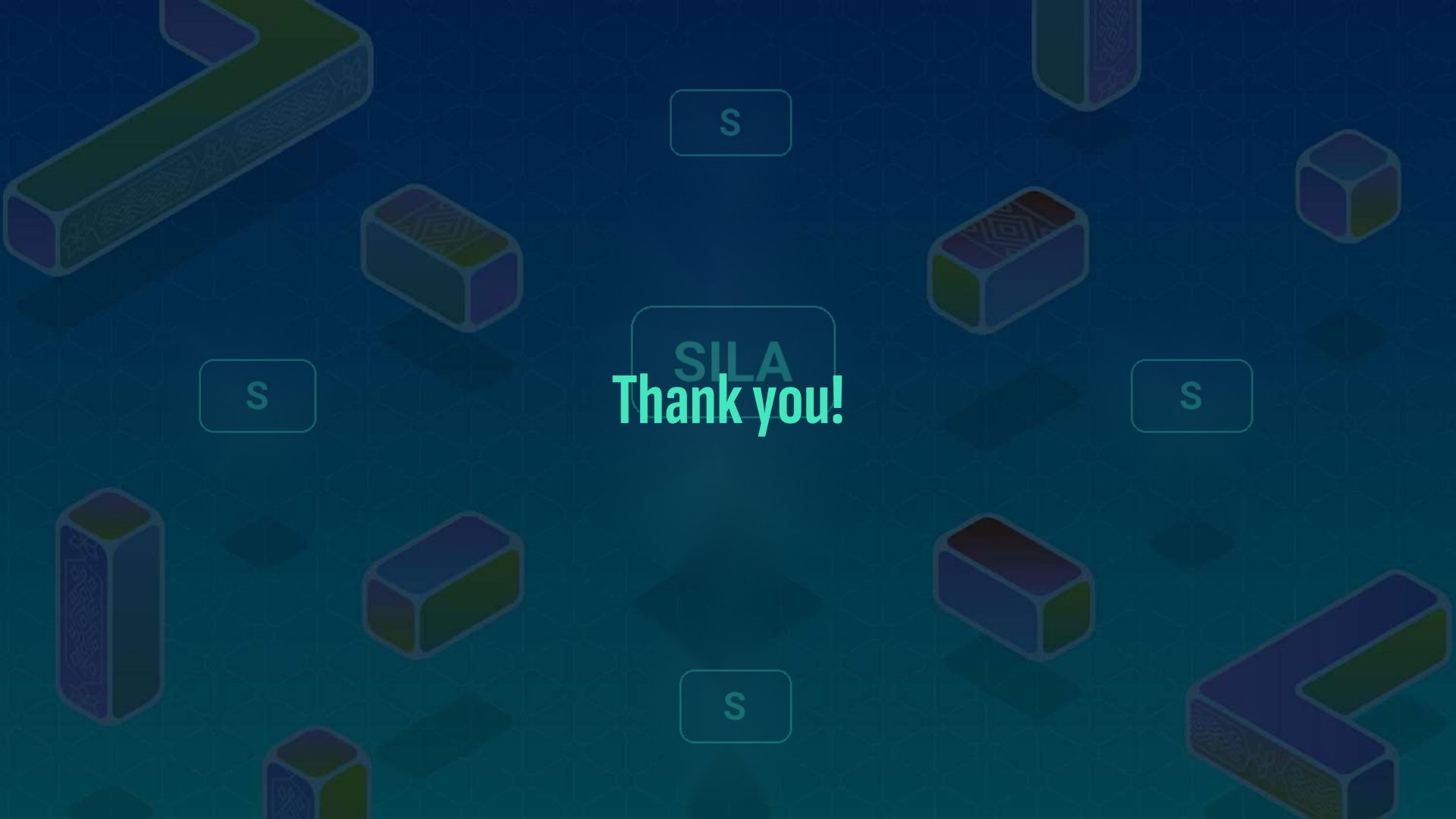
would be interesting to see how execution time changes if you went fully into eoa or fully into contract rather than

ACKNOWLEDGEMENTS

SILA
Kudos to:



NETHERMIND



SILA
Thank you!

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egestas eget quam. Donec sed odio dui.
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- Sollicitudin
- Consectetur
 - Condimentum
 - **Magna**
 - **Ligula**

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- Sollicitudin
- Consectetur
 - Condimentum
 - **Magna**
 - **Ligula**

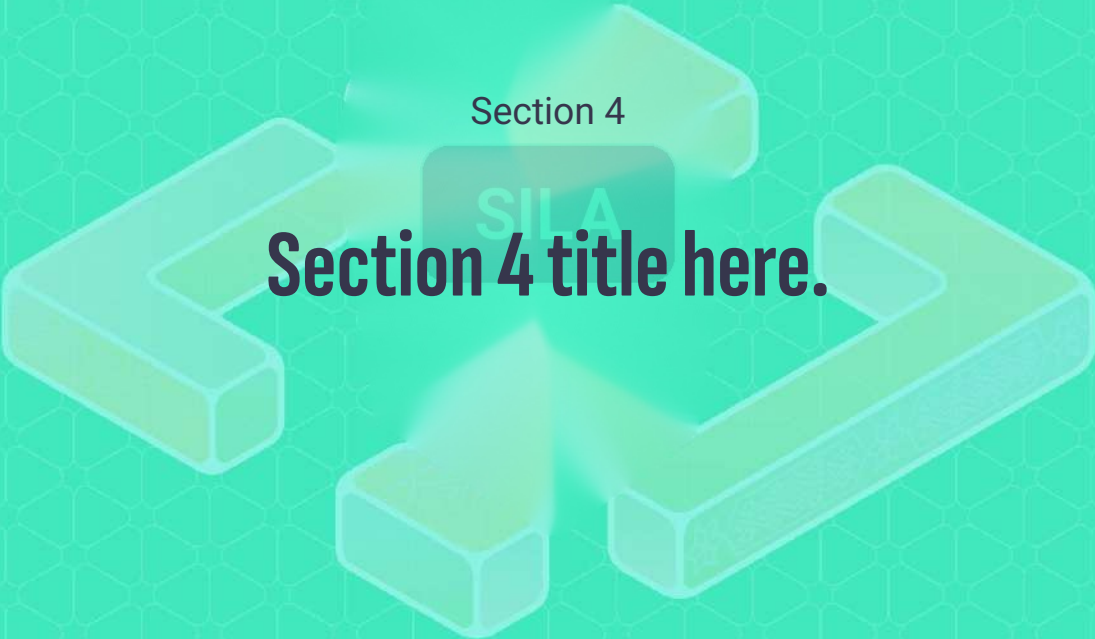
SILA

Section 2 details with a main point. Enter title here.

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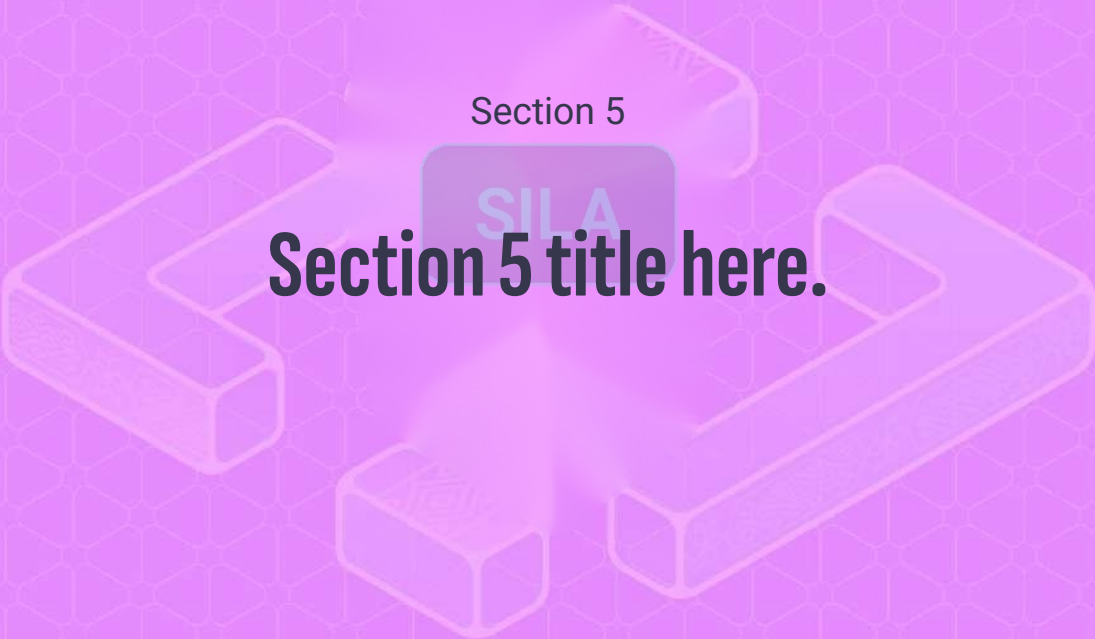
Slide

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Section 4

Section 4 title here.



Section 5

Section 5 title here.



Appendix

Some extra assets

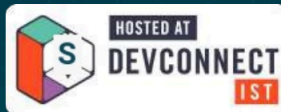


S sila
foundation



S ecosystem
support
program

S fellowship
program



Appendix

Some extra assets

